6-state curvature-Taylor model: a' driven by the curvature a''

0. Definitions

$$x[k{+}1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k], \quad a_x = \Delta t/\gamma(h), \quad \delta x_D \equiv 0$$

$$\delta x_m[k] = \delta x[k{-}d] + n_x[k], \qquad \delta x = x_d - x, \qquad h = h_d - \delta x, \qquad \Delta h_d[k] = h_d[k{+}1] - h_d[k]$$

$$\delta x[k{+}1] = \lambda_c\, \delta x[k] - F_{dx}[k]\, e_{a_x}[k] - \varepsilon[k], \qquad F_{dx} = f_{dx} + (1{-}\lambda_c)\sum_{i=1}^{d} f_{dx}[k{-}i]$$

$$\varepsilon = \delta x_T^d - \delta x_{\delta af}^d + (1{-}\lambda_c) n_x, \qquad \sigma_\varepsilon^2 := \gamma_\varepsilon(0) = \big(1 + 2(1{-}\lambda_c)^2\big)\sigma_{\delta h}^2\ (= Q_{33}), \quad \sigma_{\delta h}^2 := 4k_B T\, a_x$$

$$\mathbf{x}_e = [\,\delta x_1, \delta x_2, \delta x_3,\ a_x,\ a',\ a''\,]^\top, \qquad e_{a_x} = a_x - \hat{a}_x,\ e_{a'} = a' - \hat{a}',\ e_{a''} = a'' - \hat{a}''$$

1. Gain model (2nd-order Taylor about the desired trajectory)

$x[k{+}1] = x_d[k{+}1] - \delta x[k{+}1]$; expand a_x and a' about the desired height, drop $O(\delta x^2)$:

$$\boxed{\begin{array}{ll} a_x[k{+}1] = a_{x_{det}}[k{+}1] - a'\,\delta x[k{+}1], & a_{x_{det}}[k{+}1] = a_{x_{det}}[k] + a'\,\Delta h_d[k] \\ a'[k{+}1] = a'_{det}[k{+}1] - a''\,\delta x[k{+}1], & a'_{det}[k{+}1] = a'_{det}[k] + a''\,\Delta h_d[k] \\ a''[k{+}1] \approx a''[k] =: a'' & \end{array}}$$

$$a' = -\frac{a_x}{c}\frac{K_h}{R}, \qquad a'' = -\frac{a_x}{R^2}\big(K_h' - K_h^2\big), \qquad K_h = \frac{1}{c}\frac{dc}{d\bar{h}}, \quad K_h' = \frac{dK_h}{d\bar{h}}$$

2. True recursion

(substitute $a_{det} = a + (\text{slope}) \cdot \delta x$, then $\delta x[k{+}1]$; each level mirrors the one below)

$$\boxed{\begin{array}{l} a_x[k{+}1] = a_x + a'\,\Delta h_d + (1{-}\lambda_c) a'\,\delta x + a'\,F_{dx}\,e_{a_x} + a'\,\varepsilon \\ a'[k{+}1] = a' + a''\,\Delta h_d + (1{-}\lambda_c) a''\,\delta x + a''\,F_{dx}\,e_{a_x} + a''\,\varepsilon \\ a''[k{+}1] = a'' + a'''\,\varepsilon \qquad \text{(frozen: sweep } a'''\Delta h_d \to \text{feedforward, keep thermal } a'''\varepsilon\text{)} \end{array}}$$

3. Estimator (predict + update)

($E\{e_{a_x}\}{=}0,\ E\{\varepsilon\}{=}0$; $\delta x \to \delta\hat{x}_3$)

$$\boxed{\begin{array}{l} \delta\hat{x}_1[k{+}1] = \delta\hat{x}_2, \quad \delta\hat{x}_2[k{+}1] = \delta\hat{x}_3, \quad \delta\hat{x}_3[k{+}1] = \lambda_c\,\delta\hat{x}_3 \\ \hat{a}_x[k{+}1] = \hat{a}_x + \hat{a}'\,\Delta h_d + (1{-}\lambda_c)\hat{a}'\,\delta\hat{x}_3 \\ \hat{a}'[k{+}1] = \hat{a}' + \hat{a}''\,\Delta h_d + (1{-}\lambda_c)\hat{a}''\,\delta\hat{x}_3 \qquad \hat{a}''[k{+}1] = \hat{a}'' \end{array}}$$

$$\hat{\mathbf{x}}[k] = \hat{\mathbf{x}}[k|k{-}1] + \mathbf{L}[k]\big(\mathbf{y}[k] - \mathbf{H}\,\hat{\mathbf{x}}[k|k{-}1]\big), \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

4. Error dynamics

(True − Estimator; keep leading order, $O(a''\,\delta x)$ and up dropped)

$$\boxed{\begin{array}{l} e_{\delta x_3}[k{+}1] = \lambda_c\, e_{\delta x_3} - F_{dx}\, e_{a_x} - \varepsilon \\ e_{a_x}[k{+}1] = (1{+}a'F_{dx})\, e_{a_x} + (1{-}\lambda_c) a'\, e_{\delta x_3} + \Delta h_d\, e_{a'} + \underbrace{a'\,\varepsilon}_{q_4} \\ e_{a'}[k{+}1] = (1{+}a''F_{dx})\, e_{a'} + (1{-}\lambda_c) a''\, e_{\delta x_3} + \Delta h_d\, e_{a''} + \underbrace{a''\,\varepsilon}_{q_5} \\ e_{a''}[k{+}1] = e_{a''} + \underbrace{a'''\,\varepsilon}_{q_6} \end{array}}$$

5. $\mathbf{F}_e$ (Cols: $\delta x_1, \delta x_2, \delta x_3, a_x, a', a''$)

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx} & 0 & 0 \\ 0 & 0 & (1-\lambda_c)a' & 1+a'F_{dx} & \Delta h_d & 0 \\ 0 & 0 & (1-\lambda_c)a'' & a''F_{dx} & 1 & \Delta h_d \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}$$

Leading order (a'', a''' small): row 5 $\to [\,0\,0\,0\,0\,1\,\Delta h_d\,]$, so the hierarchy is a Δh_d-coupled chain $a_x \leftarrow a' \leftarrow a''$. In a hold Δh_d=0: $a'[k+1]$=a', a'' decoupled $\Rightarrow$ no drift (the IRW's fatal $F_e(5,6)$=1 becomes Δh_d).

6. Process noise (rank-1 thermal)

(one ε lifts through the curve derivatives)

$$\mathbf{q}[k] = \varepsilon[k]\,[\,0,\ 0,\ -1,\ a',\ a'',\ a'''\,]^\top \quad \Rightarrow \quad \mathbf{Q}[k] = \sigma_\varepsilon^2\,\hat{\mathbf{q}}\,\hat{\mathbf{q}}^\top, \quad \hat{\mathbf{q}} = [\,0, 0, -1, a', a'', a'''\,]^\top$$

$$\boxed{Q_{33} = \sigma_\varepsilon^2, \quad Q_{44} = {a'}^2\sigma_\varepsilon^2, \quad Q_{55} = {a''}^2\sigma_\varepsilon^2, \quad Q_{66} = {a'''}^2\sigma_\varepsilon^2 \qquad (\text{off-diag } Q_{ij} = \hat{q}_i\hat{q}_j\,\sigma_\varepsilon^2)}$$

$a''' = da''/d\bar{h} \cdot (1/R)$ (curve 3rd derivative). Deterministic sweeps ($a'\Delta h_d, a''\Delta h_d$) are feedforward (§3), *not* Q.

7. Initial covariance (wall-aware) & acquisition

General onset handling for the whole Taylor hierarchy: each wall-derivative state starts at its far-field value ($\to 0$) with a prior equal to its far-field magnitude $\Rightarrow$ no pent-up uncertainty released when Δh_d opens observability.

$$\boxed{P_{55,0} = [a'(\bar{h}_{init})]^2, \qquad P_{66,0} = [a''(\bar{h}_{init})]^2 \qquad (\bar{h}_{init} \text{ far} \Rightarrow\ a', a'' \to 0 \Rightarrow\ \text{pinned})}$$

c-free form: small constants at the natural scales $a' \sim a_{nom}/R$, $a'' \sim a_{nom}/R^2$ (start-far confidence). Acquisition: a', a'' estimated as states (curvature model, c-free), or evaluated from the curve (oracle, §1).